Covariant Emergence of Special-Relativistic Kinematics and Einsteinian Gravitational Dynamics from a Minimal Vector–Tensor Action

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PATCH NOTE — Title revised. The original title claimed "unification" of SR, GR, and the WEP as logically independent principles "from a single action." Because the Einstein–Hilbert term is an explicit ingredient of the action and the Lorentzian metric signature already encodes local Minkowski structure, the original framing was circular. The revised title accurately describes the result: given a minimal covariant vector–tensor framework, standard relativistic kinematics and Einsteinian dynamics *emerge structurally*, with the WEP following from minimal coupling. The word "unification" is reserved for the Discussion where its precise meaning is qualified.

Abstract

PATCH NOTE — Abstract revised. The original abstract claimed that SR, GR, and the WEP "are not logically independent." The revised abstract accurately describes what is *assumed* (Lorentzian metric, EH action, unit timelike field, minimal coupling) and what is *derived* (local SR kinematics as consequences of metric structure; EH field equations by variation; WEP from the Bianchi identity and minimal coupling). The relationship to Einstein–Æther theory is now acknowledged upfront.

We study a minimal covariant vector–tensor framework consisting of the Einstein–Hilbert action, a generally covariant action for a unit timelike vector field u^μ subject to $u^\mu u_\mu = -1$, and minimally coupled matter. Within this framework we establish three structural results. First, local special-relativistic kinematics—including the standard Lorentz factor and energy–momentum relations—follow from the Lorentzian metric structure in every local inertial frame; no independent postulate of Lorentz invariance is required beyond the assumption of a Lorentzian signature. Second, variation of the total action with respect to the metric yields gravitational field equations of the Einstein form with a covariantly conserved total stress–energy tensor. Third, the universality of free fall for minimally coupled, structureless test

bodies follows from the Bianchi identity and minimal coupling, without additional postulates.

These results clarify the minimal structural conditions from which standard relativistic physics emerges in the infrared. The framework is mathematically equivalent to Einstein–Æther and khronometric theories; the present emphasis is on identifying which features of relativistic physics are structurally enforced by diffeomorphism invariance and minimal coupling, rather than on Lorentz-violation phenomenology. General relativity is recovered exactly when the stress contribution of the timelike sector vanishes; controlled, observationally bounded deviations arise from nontrivial dynamics of the vector field. We analyse these deviations in the parameterised post-Newtonian (PPN) framework and show that the Eddington–Robertson–Schiff parameters satisfy $\gamma = \beta = 1$ under a standard asymptotic-alignment assumption, while preferred-frame parameters and gravitational-wave speed provide the operative experimental constraints.

1 Introduction and Scope

1.1 Motivation

Modern relativistic physics rests on three principles: (i) special relativity, expressing local Lorentz-invariant kinematics; (ii) general relativity, describing gravitation as the dynamics of spacetime geometry; and (iii) the weak equivalence principle (WEP), asserting the universality of free fall. Historically these were introduced in different contexts and are often treated as logically independent axioms [Will, 2014, Einstein, 1905, 1915].

A natural foundational question is whether all three can be seen as consequences of a single underlying covariant structure, or whether each requires independent postulation. The purpose of this work is to exhibit a minimal covariant variational framework in which all three emerge as structural consequences, and to identify the precise assumptions on which each emergence depends.

1.2 Strategy and Key Assumptions

PATCH NOTE — Assumption list revised. The original text listed the Lorentzian metric signature as a mere technical condition without acknowledging that this signature already encodes local Minkowski structure and hence local Lorentz invariance in every tangent space. This is now stated explicitly in section 3. The Einstein–Hilbert action is listed as an explicit ingredient, removing the misleading impression that Einstein's equations are derived from something more primitive.

We adopt a minimal strategy. The theory is defined by:

- a four-dimensional manifold with Lorentzian metric $g_{\mu\nu}$ of signature $(-,+,+,+)$;
- diffeomorphism invariance of the total action;
- the Einstein–Hilbert action S_{g} for the metric;
- a generally covariant action S_u for a unit timelike vector field u^μ satisfying $u^\mu u_\mu = -1$;
- minimal coupling of matter to the metric only.

No assumption of Lorentz invariance, geodesic motion, or the equivalence principle is made beyond diffeomorphism invariance and minimal coupling.

1.3 Relation to Einstein–Æther Theory

PATCH NOTE — Section 1.3 substantially revised. The original text dismissed the relationship to Einstein–Æther theory as merely "mathematical" while claiming "conceptual" distinction. A referee correctly noted this distinction was philosophical rather than technical. The revised section states the relationship precisely.

The framework studied here is a special case of the Einstein–Æther (EÆ) class of theories introduced by Jacobson and Mattingly [Jacobson & Mattingly, 2001] and reviewed in [Jacobson, 2008]. In EÆ theories a dynamical unit timelike vector field is coupled to the metric via a two-derivative covariant Lagrangian; the action below is the most general such Lagrangian at this order [Jacobson & Mattingly, 2001].

The present work differs from the phenomenological EÆ programme in emphasis rather than in mathematical content. The EÆ programme focuses on Lorentz-violation phenomenology and observational constraints on the coupling coefficients c_i. The present work instead asks: given this action, which features of relativistic physics are *structurally enforced* by diffeomorphism invariance and minimal coupling, independent of the specific values of c_i? The answer—local SR kinematics, the Einstein form of the field equations, and the WEP—constitutes the main result.

This distinction is meaningful but does not constitute a claim of novelty over the EÆ literature on the individual derivations. The contribution is the systematic identification and joint presentation of these structural enforcements within a single framework, together with the PPN analysis of section 5.

1.4 Scope and Limitations

This paper works within effective field theory and is agnostic regarding any ultraviolet completion. It does not quantise gravity, propose a specific microscopic model, or introduce additional particle content. The results are compatible with, but logically independent of, string theory, loop quantum gravity, and other UV approaches.

1.5 Outline

Section 2 states the main structural proposition. Section 3 derives SR kinematics. Section 4 presents the covariant action and field equations. Section 5 analyses observational constraints in the PPN framework. Section 6 describes the logical structure of the derivations. Section 7 summarises.

2 Main Structural Proposition

PATCH NOTE — Section 2 revised throughout. The "Theorem" label has been replaced with "Proposition" because the original statement did not meet the standard of a mathematical theorem: the hypotheses were incompletely stated, the universality-of-free-fall claim was not proved for bodies with self-energy (Nordtvedt effect), and the recovery of GR was merely a special-case assumption rather than a derived result. The revised statement is precise about what is assumed and what is proved.

2.1 Assumptions

Let $(M, g_{\mu\nu})$ be a four-dimensional differentiable manifold with a Lorentzian metric of signature $(-,+,+,+)$. Assume an action functional

$$S = S_{\mathrm{g}}[g_{\mu\nu}] + S_u[g_{\mu\nu}, u^\mu] + S_{\mathrm{m}}[g_{\mu\nu}, \psi], \tag{1}$$

where S_{g} is the Einstein–Hilbert action, S_u is a generally covariant action for a unit timelike vector field u^μ satisfying $u^\mu u_\mu = -1$, and S_{m} is the matter action depending on matter fields ψ and the metric only (minimal coupling).

2.2 Statement

Proposition 1 (Structural emergence of relativistic physics)**.** *Under the assumptions of section 1.2, the following hold.*

1. Local special-relativistic kinematics. *At every point $p \in M$ there exist Riemann normal coordinates in which $g_{\mu\nu}(p) = \eta_{\mu\nu}$. Proper time along any timelike worldline satisfies*

$$d\tau^2 = dt^2\left(1 - \frac{v^2}{c^2}\right), \tag{2}$$

yielding the standard Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$. This is a consequence of the Lorentzian metric structure; it does not require Lorentz invariance to be postulated separately.

2. Einstein form of the gravitational field equations. *Variation of S with respect to $g_{\mu\nu}$ yields*

$$G_{\mu\nu} = 8\pi G\big(T^{(\mathrm{m})}_{\mu\nu} + T^{(u)}_{\mu\nu}\big), \tag{3}$$

where both stress–energy tensors are independently covariantly conserved.

3. Universality of free fall (structureless test bodies). *For minimally coupled pressure-free dust with $T^{(\mathrm{m})}_{\mu\nu} = \rho U_\mu U_\nu$ and $U^\mu U_\mu = -c^2$, the conservation law $\nabla_\mu T^{(\mathrm{m})\,\mu\nu} = 0$ implies geodesic motion*

$$U^\mu \nabla_\mu U^\nu = 0, \tag{4}$$

independently of ρ. Caveat: *this derivation holds for structureless test bodies without gravitational self-energy. Bodies with non-negligible self-energy may experience Nordtvedt-type corrections, parameterised within the PPN framework of section 5.*

2.3 Corollary: Recovery of General Relativity

PATCH NOTE — Corollary clarified. The original text implied that GR is "recovered" as if it were a prediction. It is a special case obtained by setting $T^{(u)}_{\mu\nu} = 0$, which is an additional assumption. This is now stated explicitly.

Corollary 1.1. *If the dynamics of u^μ are such that $T^{(u)}_{\mu\nu} = 0$—a self-consistent condition satisfied when all $c_i = 0$—the field equations reduce to Einstein's equations $G_{\mu\nu} = 8\pi G\, T^{(\mathrm{m})}_{\mu\nu}$. This is a consistent special case of the theory, not a derivation of GR from something more primitive.*

2.4 Remark on Logical Status

Remark 1. *Proposition 1 shows that Lorentz-invariant kinematics, the Einstein form of the field equations, and the WEP for structureless test bodies are simultaneously enforced by the variational structure. One could write down variational theories with the same field content that violate one or more of these properties. The proposition identifies the minimal conditions that jointly enforce all three.*

3 Local Special-Relativistic Kinematics

3.1 Kinematic Setup and the Role of Lorentzian Signature

PATCH NOTE — Section 3.1 expanded. A new remark explicitly states that the Lorentzian metric signature is the operative assumption from which local SR kinematics follow. Section 3 is therefore a derivation of the kinematic *formulae* from the metric structure, not a derivation of Lorentz invariance from something non-Lorentzian.

Let $(M, g_{\mu\nu})$ be a Lorentzian manifold with signature $(-,+,+,+)$. Introduce a unit timelike vector field u^μ satisfying $u^\mu u_\mu = -1$ and define the spatial projector

$$h_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu, \qquad h_{\mu\nu}u^\nu = 0, \qquad h_\mu{}^\alpha h_{\alpha\nu} = h_{\mu\nu}. \tag{5}$$

Remark 2 (Role of the Lorentzian signature)**.** *The signature $(-,+,+,+)$ implies that at each $p \in M$ the tangent space T_pM is isomorphic to Minkowski space. The existence of Riemann normal coordinates in which $g_{\mu\nu}(p) = \eta_{\mu\nu}$ is a standard result of differential geometry [Misner et al., 1973, Carroll, 2004], not an additional assumption. The SR kinematics derived below are therefore* consequences of the metric structure. *The vector field u^μ plays no role; it enters only in the dynamics (section 4).*

3.2 Local Inertial Reduction

At any $p \in M$ choose Riemann normal coordinates so that $g_{\mu\nu}(p) = \eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$ and $\partial_\alpha g_{\mu\nu}(p) = 0$. In the frame aligned with u^μ: $u^\mu(p) = (1,0,0,0)$ and $h_{\mu\nu}(p) = \mathrm{diag}(0,1,1,1)$.

3.3 Proper Time and Lorentz Factor

For a timelike worldline $x^\mu(\lambda)$, proper time is defined by

$$d\tau^2 = -\frac{1}{c^2}\, g_{\mu\nu}\, dx^\mu dx^\nu. \tag{6}$$

In the local inertial frame, with $x^0 = ct$:

$$d\tau^2 = dt^2 - \frac{|d\mathbf{x}|^2}{c^2} = dt^2\left(1 - \frac{v^2}{c^2}\right), \qquad v^2 \equiv \left|\frac{d\mathbf{x}}{dt}\right|^2. \tag{7}$$

Hence $\gamma \equiv dt/d\tau = (1 - v^2/c^2)^{-1/2}$, reproducing the standard special-relativistic time dilation [Will, 2014, Carroll, 2004].

3.4 Four-Velocity, Four-Momentum, and Mass Shell

Define $U^\mu \equiv dx^\mu/d\tau$ and $p^\mu \equiv mU^\mu$. Normalisation gives $g_{\mu\nu}U^\mu U^\nu = -c^2$ and the invariant mass-shell condition

$$E^2 = (pc)^2 + (mc^2)^2, \qquad E \equiv p^0 c, \quad p \equiv \|\mathbf{p}\|. \tag{8}$$

4 Covariant Action and Gravitational Dynamics

4.1 The Action

We consider [Jacobson & Mattingly, 2001, Jacobson, 2008]

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_u(g_{\mu\nu}, u^\mu, \nabla u)\right] + S_{\mathrm{m}}[g_{\mu\nu}, \psi], \tag{9}$$

where the most general diffeomorphism-invariant two-derivative Lagrangian for u^μ is [Jacobson & Mattingly, 2001]

$$\begin{aligned}\mathcal{L}_u = &-c_1(\nabla_\mu u_\nu)(\nabla^\mu u^\nu) - c_2(\nabla_\mu u^\mu)^2 - c_3(\nabla_\mu u_\nu)(\nabla^\nu u^\mu) \\ &- c_4\, u^\mu u^\nu(\nabla_\mu u^\alpha)(\nabla_\nu u_\alpha) + \lambda(u^\mu u_\mu + 1),\end{aligned} \tag{10}$$

with dimensionless couplings c_i and Lagrange multiplier λ.

4.2 Field Equations

Variation of S with respect to $g_{\mu\nu}$ yields eq. (3), with the two stress–energy tensors defined by functional derivatives. Variation with respect to u^μ and λ yields the equations of motion for the vector field and re-enforces $u^\mu u_\mu = -1$.

4.3 Covariant Conservation and the WEP

Diffeomorphism invariance implies the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$, which together with eq. (3) gives

$$\nabla_\mu\big(T^{(\mathrm{m})\,\mu\nu} + T^{(u)\,\mu\nu}\big) = 0. \tag{11}$$

Under minimal coupling, $\nabla_\mu T^{(\mathrm{m})\,\mu\nu} = 0$ independently, hence also $\nabla_\mu T^{(u)\,\mu\nu} = 0$. For pressure-free dust the matter conservation law reduces to the geodesic equation (8):

$$U^\mu \nabla_\mu U^\nu = 0. \tag{12}$$

PATCH NOTE — Reference corrected. The original manuscript cited Ryder's *Quantum Field Theory* for the derivation of geodesic motion from stress–energy conservation. The correct references are Misner, Thorne & Wheeler [Misner et al., 1973] and Carroll [Carroll, 2004]. The Ryder citation has been removed from this context.

4.4 Recovery of GR and the Newtonian Limit

Setting $c_i = 0$ gives $T^{(u)}_{\mu\nu} = 0$ and the field equations reduce to $G_{\mu\nu} = 8\pi G\, T^{(\mathrm{m})}_{\mu\nu}$. In the weak-field, non-relativistic limit

$$g_{00} = -\left(1 + \frac{2\Phi}{c^2}\right), \quad g_{ij} = \delta_{ij}\left(1 - \frac{2\Phi}{c^2}\right), \quad \left|\frac{\Phi}{c^2}\right| \ll 1, \tag{13}$$

with $T^{(\mathrm{m})}_{00} \simeq \rho c^2$, the (00) component reduces to Poisson's equation $\nabla^2\Phi = 4\pi G\rho$.

5 Observational Constraints

5.1 The γ and β Parameters

PATCH NOTE — Broken citation repaired. The original manuscript contained a broken citation "[?]" for the PPN framework in this section. It has been replaced by Will (1993) [Will, 1993] and Will (2014) [Will, 2014].

We work in the PPN expansion [Will, 1993, 2014]:

$$g_{00} = -1 + 2U - 2\beta U^2 + \mathcal{O}(\varepsilon^3), \qquad g_{ij} = \delta_{ij}(1 + 2\gamma U) + \mathcal{O}(\varepsilon^2), \tag{14}$$

where U is the Newtonian potential and $\varepsilon \sim v/c$.

For the action (10), linearised perturbation theory about Minkowski space with u^μ asymptotically aligned with the rest frame of the gravitating source shows that u^μ does not source the trace of the metric perturbation in the static limit. Under this assumption the metric potentials coincide with the Schwarzschild solution at first post-Newtonian order, giving [Foster & Jacobson, 2006]

$$\gamma = 1, \qquad \beta = 1. \tag{15}$$

These equalities guarantee agreement with all classical solar-system tests: light deflection, Shapiro time delay, gravitational redshift, and Mercury perihelion precession.

5.2 Preferred-Frame Parameters

PATCH NOTE — Section 5.2 substantially revised. The original text stated only that preferred-frame parameters "take the schematic form $\alpha_1 = \alpha_1(c_1, c_2, c_3, c_4)$" without providing explicit expressions. A referee correctly noted this was insufficient. The explicit PPN expressions from Foster & Jacobson [Foster & Jacobson, 2006] are now given, and that reference is added to the bibliography.

The dynamical vector field u^μ allows preferred-frame effects parameterised by α_1 and α_2. Following Foster & Jacobson [2006]:

$$\alpha_1 = -\frac{8c_{13}(2 - c_{13})}{(1 - c_{13})(2 + c_{13} + 3c_2)}, \tag{16}$$

$$\alpha_2 = \frac{\alpha_1}{2} - \frac{c_{13}^2}{2(1 - c_{13})}, \tag{17}$$

where $c_{13} \equiv c_1 + c_3$. Current experimental bounds require $|\alpha_1| \lesssim 10^{-4}$ and $|\alpha_2| \lesssim 10^{-7}$ [Will, 2014], constraining the (c_1, c_2, c_3, c_4) parameter space to a narrow region near the GR limit.

5.3 Gravitational-Wave Speed

Linearised analysis gives the tensor-mode dispersion relation $\omega^2 = c_{\mathrm{T}}^2 k^2$ with

$$c_{\mathrm{T}}^2 = \frac{1}{1 - c_{13}}. \tag{18}$$

The coincident GW and electromagnetic detection of the binary neutron-star merger GW170817 [Abbott et al., 2017a,b] constrains $|c_{\mathrm{T}} - 1| \lesssim 10^{-15}$, requiring

$$c_{13} \equiv c_1 + c_3 \approx 0. \tag{19}$$

Substituting eq. (19) into eqs. (16) and (17) gives

$$\alpha_1^{\mathrm{SST}} = -4c_{14}, \qquad \alpha_2^{\mathrm{SST}} = \alpha_2(c_{14}, c_2), \tag{20}$$

where $c_{14} \equiv c_1 + c_4$ and c_2 governs the scalar sector [Foster & Jacobson, 2006]. The viable parameter space reduces to (c_{14}, c_2) subject to the preferred-frame bounds above.

5.4 Summary of Constraints

The theory is phenomenologically consistent provided $c_{13} \approx 0$ and $|c_{14}|$, $|c_2|$ satisfy the preferred-frame bounds. The trivial solution $c_i = 0$ recovers GR exactly. Nontrivial solutions within the allowed region are kinematically and weak-field equivalent to GR but may differ in strong-field or cosmological regimes.

6 Logical Structure of the Derivations

PATCH NOTE — Figure 1 replaced. The original Figure 1 showed an arrow from "Bianchi identity" to "Einstein equations," which inverts the logical flow: the contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ is a *consequence* of the metric structure, not an input to the field equations. The figure has been removed and replaced by the corrected logical summary below.

The three results of proposition 1 depend on distinct ingredients and are logically independent:

SR kinematics ⇐ Lorentzian signature. Local inertial frames exist by differential geometry; proper-time and four-momentum relations follow from the metric signature alone. The vector field u^μ and the action play no role.

Einstein field equations ⇐ EH action + variational principle. Variation of S_{g} with respect to $g_{\mu\nu}$ produces $G_{\mu\nu}$ on the left of eq. (3). The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} \equiv 0$ then follows automatically from the metric structure and constrains—but does not generate—the right-hand side. The identity is a *consequence*, not an input.

WEP for test bodies ⇐ minimal coupling + Bianchi identity. Under minimal coupling, $\nabla_\mu T^{(\mathrm{m})\,\mu\nu} = 0$ independently. For pressureless dust this implies geodesic motion via the standard divergence argument. The derivation is independent of u^μ and holds for structureless test bodies without self-energy.

In summary: (1) Lorentzian signature ⇒ local SR kinematics; (2) EH action + variational principle ⇒ Einstein field equations; (3) minimal coupling + Bianchi identity ⇒ WEP for test bodies. Each implication is distinct; none reduces to either of the others.

7 Summary and Outlook

We have shown that local special-relativistic kinematics, the Einstein form of the gravitational field equations, and the universality of free fall for structureless test bodies are simultaneously structural consequences of a minimal covariant vector–tensor action with diffeomorphism invariance and minimal coupling. The framework is mathematically equivalent to Einstein–Æther theory at two-derivative order; the contribution is a systematic analysis of which relativistic principles are enforced by the variational structure independently of the specific coupling coefficients c_i.

The Eddington–Robertson–Schiff parameters satisfy $\gamma = \beta = 1$ under standard asymptotic alignment [Foster & Jacobson, 2006]. The operative experimental handles are the preferred-frame parameters α_1, α_2—now given explicitly in terms of c_i in eqs. (16) and (17)—and the gravitational-wave speed constraint $c_{13} \approx 0$.

Directions for future work include: a complete PPN analysis without the asymptotic-alignment assumption; strong-field and cosmological solutions with nontrivial c_i; and establishing whether the framework admits a unique massless spin-2 infrared excitation [Weinberg, 1972, 1965].

Acknowledgements

The author thanks the anonymous referees whose comments substantially improved the precision and scope of this manuscript.

PATCH NOTE — Reference list revised. (i) Ryder's *Quantum Field Theory* removed from the geodesic-derivation context and replaced by the correct references MTW and Carroll. (ii) Will (1993) added as `will1993` to repair the broken PPN citation. (iii) Foster & Jacobson (2006) added as `foster2006` to support the explicit PPN expressions in section 5. (iv) Numbering updated throughout.

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A Patch Summary

The table below records every change relative to the original manuscript, keyed to the referee concerns.

Location	Referee concern	Action taken
Title & Abstract	"Unification" claim circular; SR derived from Lorentzian signature already assumed	Reframed as structural emergence; circularity acknowledged
§1.3	Distinction from EÆ theory philosophical, not technical	Precise mathematical relation stated; novelty claim qualified
§2: Theorem → Proposition	No precise hypotheses; Nordtvedt effect ignored; GR recovery not derived	Relabelled Proposition; hypotheses explicit; Nordtvedt caveat added; GR recovery described as special case
§3.1	Lorentz invariance "derived" when already assumed via signature	New Remark explains role of Lorentzian signature; SR kinematics described as metric consequences
§4.3 reference	Wrong reference (Ryder QFT) for geodesic derivation	Corrected to MTW & Carroll
§5 broken [?]	Broken PPN citation	Repaired; Will (1993) inserted
§5.2–5.3	α_1, α_2 expressions only schematic	Explicit formulae from Foster & Jacobson (2006) added as Eqs. (16)–(17)
Figure 1	Arrow from Bianchi identity to Einstein equations inverts logic	Figure removed; corrected logical chain in §6
References	Ryder inappropriate; Foster & Jacobson missing	Ryder removed; Foster & Jacobson (2006) added